

Ames Housing Price Prediction

EDA, Feature Engineering & Model Evaluation Report

1. Dataset & Preprocessing

The dataset is the Ames, Iowa housing dataset (Kaggle 'House Prices' train set): 1,460 rows and 81 columns (38 numerical, 43 categorical), with target variable **SalePrice**. No duplicate rows were found, but missing values were extensive — PoolQC, MiscFeature, Alley, and Fence were missing in 80–99% of rows, while GarageType/Qual/Cond/Finish and basement fields were missing in ~2–6% of rows.

Cleaning & imputation

- **Outlier removal:** Dropped 2 rows with GrLivArea $\geq 4,000$ sq ft (the dataset author's documented recommendation) — these were atypical/partial sales that also removed the extreme TotalBsmtSF outlier.
- **Missing-value logic:** NaNs in categorical quality/type columns (e.g. GarageType, BsmtQual) were filled with the explicit label "None" (absence of a feature, not truly missing); NaNs in related numeric columns (e.g. GarageArea, BsmtFinSF1) were filled with 0.
- **Statistical imputation:** LotFrontage was imputed with the per-Neighborhood median; MSZoning with the mode per MSSubClass; Electrical/Utilities/Exterior1st/Exterior2nd/KitchenQual/SaleType with the training-set mode. All imputation maps were fit on the training split only, and a train/test split (80/20, random_state=42) was performed *before* fitting these maps to avoid data leakage.

Target transformation

SalePrice is heavily right-skewed (skew = 1.88; mean \$180,921 vs. median \$163,000). A Yeo-Johnson power transform was applied to the target (and to the most skewed continuous predictors, $|\text{skew}| > 0.75$) to make relationships more linear and stabilize variance; predictions were inverse-transformed back to dollars for evaluation.

2. Key EDA Findings

- **Skew & zero-inflation:** LotArea, MasVnrArea, 2ndFlrSF, and porch fields are strongly right-skewed with heavy zero-inflation (e.g. most homes have no 2nd floor, pool, or masonry veneer) — these became strong candidates for binary "has-X" flags rather than raw magnitudes.
- **GarageArea** shows a multi-modal distribution clustering around 1-, 2-, and 3-car capacities.
- **Ordinal quality fields** (GarageQual, BsmtCond, ExterCond, Utilities) are dominated (>90%) by a single "TA"/typical category, giving them low variance; ExterQual, BsmtQual, HeatingQC, and KitchenQual showed the most useful spread as SalePrice predictors.
- **Nominal features:** SaleType, SaleCondition, MSZoning, BldgType, and RoofStyle are heavily imbalanced (one-hot encoding would create many near-zero-variance columns); Neighborhood, Exterior1st, and Exterior2nd have high cardinality, motivating target encoding instead.
- **Correlation with SalePrice:** OverallQual and GrLivArea are the dominant linear drivers; GarageCars, GarageArea, TotalBsmtSF, and 1stFlrSF are also strongly positively correlated. YrSold/MoSold show near-zero linear correlation because their relationship is cyclical, not linear.
- **Neighborhood effects:** NoRidge, NridgHt, and StoneBr are the most expensive neighborhoods (means \$310K–\$335K); BrDale, IDOTRR, and MeadowV are the cheapest (~\$100K). NNames has the most sales (225 homes).
- **Seasonality:** June has the highest sales volume (253 sales); September has the highest average price (\$195,683) and April the lowest (\$171,503).
- **Bedrooms vs. price:** more bedrooms did not mean higher price — 5+ bedroom homes tend to be older multi-family conversions rather than luxury homes.

3. Important Data Insights & Feature Engineering

Guided by the EDA above, the following engineered features were created (leak-free, fit on training data only):

- **Binary presence flags:** HasPool, Has2ndFlr, HasFireplace, HasMasVnr, HasPorch, plus flags for sparse porch/misc columns — separates "feature exists" from "how large is it".
- **Age features:** HouseAge, RemodAge, GarageAge (relative to YrSold, clipped at 0) replace raw calendar years, which are not meaningful to linear models.
- **Aggregates:** TotalBath (full + 0.5×half, house + basement) and TotalSF (TotalBsmtSF + 1stFlrSF + 2ndFlrSF).
- **Cyclic encoding:** MoSold_sin / MoSold_cos so December and January are treated as adjacent.
- **Ordinal mapping:** quality strings (Ex/Gd/TA/Fa/Po) mapped to a 0–5 numeric scale.
- **Leak-free target encoding:** Neighborhood, MSZoning, SaleType, SaleCondition, RoofStyle, and BldgType replaced with median SalePrice computed on the training set only (fallback to global median for unseen categories).
- **Interactions/ratios:** QualxCond, OverallStructureScore (sum of 5 quality scores), BsmtUnfRatio, TotalPorchSF, IsRemodeled flag.

Multicollinearity control: A VIF (variance inflation factor) pass flagged infinite VIF for features that were exact linear combinations of others (e.g. HouseAge alongside YrSold & YearBuilt). Redundant raw columns were dropped in favor of their engineered composites (e.g. dropping OverallQual/OverallCond in favor of QualxCond and OverallStructureScore; dropping 1stFlrSF/2ndFlrSF/TotalBsmtSF in favor of TotalSF). This reduced the feature set to 84 columns, after which the top remaining VIF was TotalSF (39.8) and GrLivArea

(26.1) — expected, since living area is the strongest price driver. Remaining numeric features were scaled with RobustScaler (fit on train only); binary flags were left unscaled. The final model-ready matrix had 53 numeric features.

4. Models Used

Three regression models were trained on the Yeo-Johnson-transformed target and evaluated after inverse-transforming predictions back to dollars:

- **ElasticNet** — linear model combining L1 (Lasso) and L2 (Ridge) regularization.
- **Support Vector Regressor (SVR)** — RBF-kernel model for capturing non-linear relationships.
- **XGBoost Regressor** — gradient-boosted tree ensemble.

5. Hyperparameter Tuning

ElasticNet and SVR were tuned with 5-fold **GridSearchCV**; XGBoost was tuned with 15-iteration 5-fold **RandomizedSearchCV** (for speed over a larger space). All used neg-RMSE scoring on the transformed target.

Model	Search space	Best parameters	Best CV RMSE (transformed)
ElasticNet	alpha: [1e-4..10], l1_ratio: [0.1..0.9]	alpha=0.001, l1_ratio=0.9	0.05524
SVR	C: [0.1..1000], epsilon: [0.001..0.2], kernel=rbf	C=0.1, epsilon=0.05, kernel=rbf	0.15303
XGBoost	n_estimators, max_depth, learning_rate, subsample, colsample_bytree	n_estimators=200, max_depth=3, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8	0.05615

Interpretation: ElasticNet's high l1_ratio (0.9) indicates the optimal model leans heavily toward Lasso-style sparsity, zeroing out weak engineered features. SVR needed strong regularization (small C=0.1) to avoid overfitting a high-dimensional RBF mapping of the encoded feature space.

6. Evaluation Results

Model	Train R ²	Test R ²	Test RMSE (\$)	Test MAE (\$)
ElasticNet	0.9246	0.9194	\$20,565.07	—
XGBoost	—	—	\$20,985.77	—
SVR	0.3094	0.2345	\$63,386.41	—

(Train R² / MAE were not computed in the notebook for XGBoost and SVR's MAE at the point of writing; figures above reflect what was explicitly printed.) All metrics are on the untransformed dollar scale after inverse-transforming predictions.

ElasticNet and XGBoost performed very similarly (~\$20.6K–\$21.0K test RMSE, both explaining roughly 92% of price variance), while SVR performed far worse (R² dropped to ~0.23 on test, RMSE over \$63K). The gap between SVR's train R² (0.31) and test R² (0.23) — both low — suggests SVR was underfitting rather than overfitting once heavily regularized, and struggled to map the wide, encoded feature space onto the price surface.

A separate, simpler baseline (RandomForestRegressor on a standard ColumnTransformer pipeline, without the full custom feature-engineering pass) was also tried earlier in the notebook and reached RMSE \$25,754, MAE \$18,081, R² 0.874 — useful as a sanity check, but outperformed by the tuned ElasticNet/XGBoost pipeline built on the engineered features.

7. Final Model Selection

ElasticNet is the selected final model. It achieved the lowest test RMSE (\$20,565) and highest test R² (0.919) of the tuned candidates, narrowly ahead of XGBoost, while being simpler, faster to train, and more interpretable (linear coefficients on a carefully engineered, VIF-pruned feature set). The heavy feature engineering and multicollinearity reduction converted a non-linear pricing problem into one where a regularized linear model could match or beat a gradient-boosted ensemble — making ElasticNet the preferred choice for this dataset size and feature design.

8. Main Conclusions

- Overall quality (OverallQual) and total living/floor area (GrLivArea, TotalSF) are the dominant drivers of Ames house prices.
- Careful handling of skewness (Yeo-Johnson), zero-inflated features (binary flags), high-cardinality categoricals (target encoding), and multicollinearity (VIF pruning) mattered more than model choice: once features were well engineered, a regularized linear model matched a tuned gradient-boosted tree.
- SVR was not well suited to this high-dimensional, mixed encoded feature space and underperformed both other models substantially.
- The final ElasticNet pipeline predicts sale price with a typical error of roughly \$20.5K on a dataset with a median price of \$163K (~12.6% of median price), a strong result for a linear model on this dataset.

- The notebook was left unfinished in places (e.g. a leftover simpler RandomForest baseline at the end); next steps would include finalizing a single clean pipeline, adding MAE/train- R^2 reporting for all models consistently, and possibly stacking ElasticNet + XGBoost.

Source: eda_notfinished_.ipynb (Ames/House Prices dataset, Kaggle). Report generated from notebook code, comments, and printed outputs.